

Valerian Fourel

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SUMMARY

ML Engineer specializing in Causal AI and LLMs applied to complex real-world systems, from remote sensing to 3D vision modeling. I combine this diverse technical expertise with cross-cultural communication skills to build bridges across international teams. My goal is to deliver scalable, global solutions that solve multidisciplinary challenges.

EDUCATION

Eberhard Karls University Tubingen

Tubingen, Germany

MSc. Quantitative Data Science - Machine Learning Specialization

Module: **Deep Learning** - **Causal Machine Learning** - **Time Series Analysis** - **Psychometrics**

McGill University

Montreal, Canada

BA. Computer Science and Economics

Module: **Data Structures & Algorithms** - **Intelligent Robotics** - **Databases** - **Industrial Organizations**

WORK EXPERIENCE

University of Hamburg

Hamburg, Germany

Machine Learning Research Associate

Aug 2025 - Present

- **Modeled** text data mixtures as causal treatment effects using **PySpark** to process **10,000+** documents, optimizing training corpora for distinct LLM architectures.
- **Executed** experimental fine-tuning designs on **8+** SLM variants in PyTorch, validating architectural choices that improved the reliability of **1.7B+** parameter production models.
- Synthesized complex **Causal ML research** into actionable documentation, streamlining the **architecture design process** for diverse lab stakeholders.

Economic AI GmbH

Hamburg, Germany

Software Engineer (Contract)

Aug 2025 - Present

- Implemented **PySpark** features on **Databricks** for a **causal machine learning** codebase, **reducing runtime by 4x** and improving performance for **50K+** **data pipelines** used by 8+ engineers.
- Delivered **12+** **presentations translating complex Causal ML concepts** into accessible insights, acting as a **bridge to ensure alignment between business leaders and engineers**.
- Ensured **production-grade reliability** for professional users by implementing **150+** **rigorous unit tests** and authoring in-depth **API guides**.

Max Planck Institute for Intelligent Systems

Tubingen, Germany

Research Assistant

Aug 2023 - Aug 2025

- **Engineered** data processing and embedding workflows for textual and visual data across **50,000+** **motion sequences**, using **PyTorch** to streamline workflows, and reported progress to lab stakeholders to support research goals.
- **Standardized and tested Docker containers** for **deploying machine learning models**, **reducing deployment time by 90%** and increasing **scalability** for a team of 10+ researchers, specializing in **3D human models**.
- Researched **Few-Shot Learning** techniques on **Visual Large Language Models** to optimize emotion detection in faces, resulting in a **20% increase** in emotional recognition accuracy using **500K facial emotion images**.

Soil Science Lab - University of Tubingen

Tubingen, Germany

Data Science Intern

Jan 2023 - Aug 2023

- **Built PyTorch neural networks** to predict soil metrics, leveraging **time-series geospatial data**, achieving an **RMSE below 0.05** for sustainable agriculture insights.
- **Engineered a Python ETL pipeline** for a **10 million-point satellite dataset**, applying **Explainable AI** to validate feature extraction stability for **cross-domain transfer learning** tasks.
- **Designed** regression-based experimental frameworks using **SQL** and **Python visualization suites** (Matplotlib/Seaborn) to generate interpretable environmental insights for the research team.

SKILLS & INTERESTS

Skills: Python (PyTorch, PySpark) - Causal Inference - SQL - Docker - CI/CD & Unit Testing - Azure/Databricks

Interests: Music (1950s to 1980s) - Strategic Board Games - Functional Training

Languages: French (C2) - English (C2) - German (B2) - Korean (A2)